Final Exam

Graded

Student

Menger Wen

Total Points

38 / 50 pts

Question 1

Bisection

8 / 8 pts

- ✔ **+ 1 pt** (a) Check continuity
- ✔ **+ 2 pts** (a) Different signs on two boundaries
- ✔ **+ 1 pt** (b) Partial work
- ✔ **+ 1 pt** (b) Correct answer 2^{-n}
- ✔ **+ 1.5 pts** (c) Partial work, e.g., write down the formula of relative error
- ✔ **+ 1.5 pts** (c) Correct answer 5

Question 2

Interpolation

7 / 7 pts

- ✔ **+ 1 pt** Correct formula for divided difference
- ✔ **+ 2 pts** Correct answer for $f[x_0, x_1]$
- ✔ **+ 2 pts** Correct answer for $f[x_0]$
- ✔ **+ 2 pts** Correct answer for $f[x_1]$

+ 0 pts Incorrect or blank

Question 3

Fixed Point — 4 / 5 pts

✔ **+ 1 pt** (a) is correct

✔ **+ 2 pts** (b) is correct

+ 1 pt Partial for (b)

✔ **+ 1 pt** (c) is correct

+ 1 pt (d) is correct

+ 0 pts Wrong or blank

Question 4

Quadrature — 7 / 8 pts

4.1 Quadrature — 4 / 4 pts

✔ **+ 4 pts** Correct

+ 1 pt compute integral of $ae^x + b\sin(\pi x/2)$

+ 1 pt evaluation of $f(-1)$ and $f(0)$

+ 1 pt correct w_0

+ 1 pt correct w_1

+ 1 pt partial credit

+ 0 pts missing/no progress

4.2 Richardson — 3 / 4 pts

+ 4 pts Correct

✔ **+ 2 pts** correct expansion of $N(h/3)$ or $N(h/9)$

+ 1 pt correct equation settings to cancel the h term

+ 1 pt correct equation setting to cancel the h^2 term

✔ **+ 1 pt** correct equation for M (in terms of N_1 or N_2)

+ 1 pt partial credit (e.g. try to expand $N(h/3)$ or $N(h/9)$)

+ 0 pts missing or no relevant progress

Question 5

MC **6** / 8 pts

5.1 **(a)** **2** / 2 pts

✔ **– 0 pts** Correct

– 2 pts Incorrect

5.2 **(b)** **2** / 2 pts

✔ **– 0 pts** Correct

– 2 pts Incorrect

5.3 **(c)** **0** / 2 pts

– 0 pts Correct

✔ **– 2 pts** Incorrect

5.4 **(d)** **2** / 2 pts

✔ **– 0 pts** Correct

– 2 pts Incorrect

Question 6

ODEs **2** / 8 pts

+ 8 pts Correct

+ 1 pt a) correct truncation error expression

+ 2 pts a) correct Taylor expansion (in terms of $\partial_y f$ and $\partial_t f$)

+ 1 pt a) correct bound

✔ **+ 1 pt** a) partial credit (e.g. trying to use Taylor expansion with y)

+ 2 pts b) reasonable method

+ 2 pts b) check conditions for the method

✔ **+ 1 pt** b) partial credit

+ 0 pts missing or no progress

Question 7

MATLAB

4 / 6 pts

+ 6 pts correct

+ 3 pts (a) correct

✔ **+ 3 pts** (b) correct

✔ **+ 1 pt** a) partial

+ 2 pts a) partial

+ 1 pt b) partial

+ 2 pts b) partial

+ 0 pts incorrect

Math 128a: Final exam

Tuesday, December 16, 3:00-6:00pm

LEFT NEIGBOR:	YOUR NAME:	RIGHT NEIGBOR:
Kenneth Gonzalo Fraire Marquez 304 1476896	Jie (Menger) Wen	

STUDENT ID:

3041994738

1	2	3	4	5	6	7	total
8 pts	7 pts	5 pts	8 pts	8 pts	8 pts	6pts	50 pts

Please read the following instructions carefully:

- You may not use any books, notes, or calculators.
- Switch off any electronic devices and put them in your bag (moblie phones, tablets, etc.)
- Exam duration: **180 minutes**
- Solve all problems. Answer all problems after **carefully reading** them. Write your name on all the sheets.
- **Show all your work** and explain everything you write.
- Do not use pencils but rather pens. (else you use them at your own risk and we may not be able to read your write-up)

Good luck!

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$x = \pm\frac{1}{\sqrt{3}}$



Problem 1: Bisection [8 points]

Consider $f(x) = x^3 - x - 1$.

(a) Prove that the bisection method on $[1, 2]$ will converge to a root of $f(x)$.

Continuity: $f'(x) = 3x^2 - 1$ Exists on $[1,2] \Rightarrow f(x) \in C([1,2])$

Let $f'(x) = 0$: $x = \pm\frac{1}{\sqrt{3}} < 1$ $\therefore f(x)$ strictly increases on $[1,2]$, $f'(x) > 0$

$f(1) = 1 - 1 - 1 = -1 < 0$ $f(2) = 8 - 2 - 1 = 5 > 0$

By IVT: there is a unique root for $f(x) = 0$ on $[1,2]$

$\therefore$ The bisection method on $[1,2]$ will converge to a root of $f(x)$. Q.E.D.

(b) Find a bound on the absolute error in approximating this root after n iterations.

Denote the result after n iterations is p_n, the real root is p

$|p_n - p| \leq \frac{2-1}{2^n} = \frac{1}{2^n}$

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(c) How many iterations are necessary to ensure the relative error is at most 2^{-5}? (Your answer does not have to be exact.)

By (b): $|p_n - p| \leq \frac{1}{2^n} = \frac{1}{2^5}$

$\therefore$ 5 iterations !

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Problem 2: Interpolation [7 points]

For a function f, the divided differences are given by

$x_0 = 0.0$	$f[x_0] = m = 1$	$f[x_0, x_1] = p = 5$	
$x_1 = 0.4$	$f[x_1] = n = 3$		$f[x_0, x_1, x_2] = \frac{50}{7}$
		$f[x_1, x_2] = 10$	
$x_2 = 0.7$	$f[x_2] = 6$		

Determine the missing entries.

Let $f[x_0] = m$, $f[x_1] = n$, $f[x_0, x_1] = p$

$$f[x_0, x_1, x_2] = \frac{50}{7} = \frac{10 - p}{x_2 - x_0} = \frac{10 - p}{0.7} = \frac{100 - 10p}{7} \quad \therefore p = 5$$

$$f[x_1, x_2] = 10 = \frac{6 - n}{x_2 - x_1} = \frac{6 - n}{0.3} \quad \therefore 6 - n = 3,\ n = 3$$

$$f[x_0, x_1] = p = 5 = \frac{3 - m}{x_1 - x_0} = \frac{3 - m}{0.4} \quad \therefore 3 - m = 2,\ m = 1$$

$$\therefore f[x_0] = 1,\ f[x_1] = 3,\ f[x_0, x_1] = 5$$

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$W^{-1}(p) = \frac{1}{2} = pe^p$ $\quad W^{-1}(w) = we^w$

$W(\frac{1}{2}) = p$ $\quad W^{-1}(1) = 1 \cdot e^1 = e$

$W(e) = 1$

Problem 3: Fixed points [5 points]

The Lambert W function is a function whose inverse is given by $W^{-1}(w) = we^w$. For instance, $W(e) = 1$ because $W^{-1}(1) = 1 \cdot e^1 = e$. When z is a positive real number, it is known that $W(z)$ is also a positive real number. One can compute $W(1/2)$ with the bisection method or fixed-point iteration.

(a) Write down a function $f(w)$ for which $W(1/2)$ is a fixed point.

For fixed point iteration: $f(w) = w$, for $W(\frac{1}{2})$ is a fixed point

$f(W(\frac{1}{2})) = W(\frac{1}{2})$ Let $W(\frac{1}{2}) = p$, $W^{-1}(p) = pe^p = \frac{1}{2}$ $\therefore p = \frac{1}{2e^p} = f(p)$

$\therefore f(x) = \frac{1}{2e^x}$ / $f(w) = \frac{1}{2e^w}$

$+ \frac{1}{(n+1)!} H^{(n+1)}(\xi(x)) x^{n+1}$

(b) Give a number α for which the convergence rate of fixed-point iteration is $O(\alpha^n)$.

Taylor expansion: let $H(x) = e^x = H(0) + H'(0)x + \frac{1}{2}H''(0)x^2 + \cdots + \frac{1}{n!}H^{(n)}(0)x^n$

$= 1 + x + \frac{1}{2}x^2 + \cdots + \frac{1}{n!}x^n + \frac{1}{(n+1)!}H^{(n+1)}(\xi(x))x^{n+1}$

$\therefore f(x) = \frac{1}{2} \frac{1}{1 + x + \frac{1}{2}x^2 + \cdots}$ $\quad \therefore \alpha = \frac{1}{2}$

(c) Write down a function $g(w)$ for which $W(1/2)$ is a root.

$W(\frac{1}{2})$ is a root: $g(W(\frac{1}{2})) = g(p) = 0$ $\quad W^{-1}(p) = pe^p = \frac{1}{2}$ $\quad (p = W(\frac{1}{2})$, from (a))

$\therefore g(x) = e^x - \frac{1}{2x}$ / $g(w) = e^w - \frac{1}{2w}$ ← same

Or: $g(x) = 2xe^x - 1$ / $g(w) = 2we^w - 1$ ← same

Or: $g(x) = 2e^x - \frac{1}{x}$ / $g(w) = 2e^w - \frac{1}{w}$ ←

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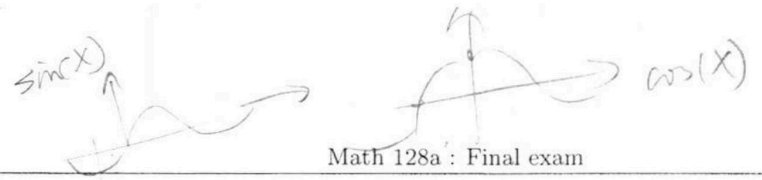
(d) Give a number β for which the convergence rate of the bisection method is $O(\beta^n)$.

By the Taylor expansion showed in (b): $e^x = 1 + x + \frac{1}{2}x^2 + \cdots$

$e^x - \frac{1}{2^x} = 1 + x + \frac{1}{2}x^2 + \cdots - \frac{1}{2} - x = \frac{1}{2} + \frac{1}{2}x^2 + \cdots \quad \therefore \beta = 1$

$2xe^x - 1 = 2x(1 + x + \frac{1}{2}x^2 + \cdots) = 2x + 2x^2 + x^3 + \cdots \quad \therefore \beta = 2$

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Problem 4: Quadrature [8 points]

(a) Find a quadrature rule

$$\widehat{I}_{[-1,0]}(f) = w_0 f(-1) + w_1 f(0),$$

that approximates the integral

$$I_{[-1,0]}(f) = \int_{-1}^{0} f(x)dx,$$

and that is exact for all functions of the form $f(x) = ae^x + b\sin(\pi x/2)$; $a, b \in \mathbb{R}$.

For $f_1(x) = e^x$: $I_{[-1,0]}(f_1) = \int_{-1}^{0} f_1(x)\,dx = \int_{-1}^{0} e^x dx = e^0 - e^{-1} = 1 - \frac{1}{e}$

$\widehat{I}_{[-1,0]}(f_1) = w_0 f_1(-1) + w_1 f_1(0) = \frac{w_0}{e} + w_1 = 1 - \frac{1}{e}$ $\quad \therefore w_0 + e w_1 = e - 1$ ①

For $f_2(x) = \sin(\frac{\pi}{2}x)$: $I_{[-1,0]}(f_2) = \int_{-1}^{0} f_2(x)\,dx = -\frac{2}{\pi}\int_{-1}^{0} \cos(\frac{\pi}{2}x)\,dx =$

$-\frac{2}{\pi}\left[\cos(0) - \cos(-\frac{\pi}{2})\right] = -\frac{2}{\pi}[1 - 0] = \frac{-2}{\pi}$

$\widehat{I}_{[-1,0]}(f_2) = w_0 \sin(-\frac{\pi}{2}) + w_1 \sin(0) = -w_0 = \frac{-2}{\pi}$ $\quad \therefore w_0 = \frac{2}{\pi}$ ②

Combine ① & ②: $\begin{cases} w_0 = \frac{2}{\pi} \\ w_1 = \frac{e - 1 - \frac{2}{\pi}}{e} \end{cases}$

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(b) Suppose that $N(h)$ is an approximation to M for every $h > 0$ and that

$$M = N(h) + K_1 h + K_2 h^2 + K_3 h^3 + \dots$$

for some constants K_1, K_2, K_3, ... Use the values $N(h)$, $N(h/3)$ and $N(h/9)$ to produce an $O(h^3)$ accurate approximation to M.

$$\begin{cases} M = N(h) + K_1 h + K_2 h^2 + K_3 h^3 + \cdots & ① \\ M = N(\frac{h}{3}) + K_1 \cdot \frac{h}{3} + K_2 \cdot \frac{h^2}{9} + K_3 \cdot \frac{h^3}{27} + \cdots & ② \\ M = N(\frac{h}{9}) + K_1 \cdot \frac{h}{9} + K_2 \cdot \frac{h^2}{81} + K_3 \cdot \frac{h^3}{729} + \cdots & ③ \end{cases}$$

After ①×a + ②×b + ③×c : There is an $O(h^3)$ accurate approximation to M

$$\therefore \begin{cases} a + \frac{1}{3}b + \frac{1}{9}c = 0 & ④ \\ a + \frac{1}{9}b + \frac{1}{81}c = 0 & ⑤ \end{cases} \quad ④ - ⑤ : \frac{2}{9}b + \frac{8}{81}c = 0 \quad \therefore \text{Let } \begin{cases} a = 1 \\ b = -4 \\ c = 9 \end{cases} ⑥$$

$1 - 4 + 9 = 6$ $\quad \therefore 6M = N(1) - 4N(\frac{h}{3}) + 9N(\frac{h}{9}) + (1 - \frac{4}{27} + \frac{1}{81})K_3 h^3 + \cdots$

$$\therefore M = \frac{1}{6}\left[N(1) - 4N(\tfrac{h}{3}) + 9N(\tfrac{h}{9})\right] + \frac{35}{243}K_3 h^3 + \cdots$$

is an $O(h^3)$ accurate approximation to M!

$1 \times ① - 4 \times ② + 9 \times ③$

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Problem 5: Multiple choice [8 points]

(a) Consider the following fixed point problem

$$p = g(p) := \frac{1}{3}p^2 + \frac{2}{3}.$$

What order of convergence can you expect for the corresponding fixed point iteration starting with starting value $p_0 = 1.5$?

- ☐ 0
- ☑ 1
- ☐ 2
- ☐ 3
- ☐ 4

(b) Assume that we have a floating point number system in which we use 1 bit for the sign s, 3 bits for the characteristic c (we take $c_0 = 3$) and 2 bits for the mantissa f. So a number is represented as

$$x = \begin{cases} (-1)^s 2^{c-3}(1+f), & 1 \le c \le 6, \\ (-1)^s 2^{-2} f, & c = 0, \\ \pm\infty, & c = 7, f = 0, \\ \text{NaN}, & c = 7, f \neq 0, \end{cases}$$

For example, the floating-point number 1.0 is represented by the bit string

0 011 00.

Find the number encoded by the following bit pattern: 1 101 11

- ☐ 2
- ☐ -1.75
- ☑ -7
- ☐ 14
- ☐ -14

(c) Suppose $p_n \to p$ quadratically ($\lim_{n\to\infty} \frac{|p_{n+1}-p|}{|p_n-p|^\alpha} = \lambda \in (0,\infty)$ with $\alpha = 2$.) Let $q_n = p_{2n}$ for $n = 1, 2, \ldots$. At what order α does q_n converge to p?

- ☐ 0
- ☐ 1
- ☑ 2
- ☐ 3
- ☐ 4

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(d) Suppose $y(t)$ satisfies $y'(t) = \sin y(t)$. Express $y'''(t)$ in terms of $y(t)$.

- ☐ $y'''(t) = -y(t)$
- ☐ $y'''(t) = \sin y(t)$
- ☐ $y'''(t) = \sin y(t) \cos y(t)$
- ☐ $y'''(t) = \cos y(t) - 2(\sin y(t))^2$
- ☐ $y'''(t) = \sin y(t) - 2(\sin y(t))^2$
- ☑ $y'''(t) = \sin y(t) - 2(\sin y(t))^3$
- ☐ $y'''(t) = \sin y(t) - 5(\sin y(t))^3$

$$y''(t) = \frac{dy'}{dt}(t) = \frac{df}{dt}(t, y(t)) = \frac{\partial f}{\partial t}(t, y(t)) + \frac{\partial f}{\partial y}(t, y(t)) \cdot f(t, y(t))$$

$$f(t, y(t)) = \sin(y(t)) = y'(t)$$

$$(\sin y(t))^2$$

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Problem 6: ODEs [8 points]

We consider the ordinary differential equation

$$y'(t) = f(t, y(t)), \quad y(0) = \alpha.$$

Assume that $f \in C^1([0,T] \times \mathbb{R})$ with bounded derivatives (and Lipschitz). for $t \in [0,T]$ and the implicit/backward Euler method

$$y_{i+1} = y_i + hf(t_{i+1}, y_{i+1}), \quad y_0 = \alpha,$$

where $h = T/N$ for some $N \in \mathbb{N}$ and $t_i = hi$, $i = 0, 1, \ldots, N$ and y_i is the approximation of $y(t_i)$ computed by the backward Euler method.

(a) Compute a bound on the truncation error for the backward Euler method.

on $[0,T] \times \mathbb{R}$

$\because f = f(t,y) \in C^1([0,T] \times \mathbb{R})$ & Lipschitz continuous in y with Lipschitz constant L

Suppose $\exists M$ with $|y''(t)| \le M$ on $[0,T]$ (y solves ODE: $y' = f(t,y)$, $y(0) = \alpha$)

Then the approximations $y_0, y_1, \cdots, y_N$, $N \in \mathbb{N}$ generated by Euler's method satisfy

$\forall i = 0, 1, 2, \cdots N$, $|y(t_i) - w_i| \le \frac{hM}{2L}\left(e^{Lt_i} - 1\right) = \frac{\frac{T}{N}M}{2L}\left(e^{Lt_i} - 1\right)$

Lemma ①: $\forall x \ge -1$, $m > 0$, $0 \le (1+x)^m \le e^{mx}$

Lemma ②: Let $s, t > 0$, $\{a_i\}_{i=0}^{k}$ sequence with $a_0 \ge -\frac{t}{s}$ & $a_{i+1} \le (1+s)a_i + t$ ⇒

$$a_{i+1} \le e^{(i+1)s}\left(a_0 + \frac{t}{s}\right) - \frac{t}{s}$$

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(b) Given y_i, what method would you suggest to find y_{i+1} for a general f? Under which conditions on f and h does the method converge?

I suggest using the Modified Euler/explicit trapezoidal rule:

$$\begin{cases} K_1 = f(t_i, y_i) \\ K_2 = f(t_i + h, y_i + hK_1) \\ y_{i+1} = y_i + \frac{h}{2}(K_1 + K_2) \end{cases}$$

Conditions: ① $f = f(t,y) \in C([0,T]\times\mathbb{R})$ & Lipschitz continuous in y with Lipschitz constant L on $[0,T]\times\mathbb{R}$

② $\exists M$ with $|y''(t)| \le M$ on $[0,T]$ (y solves ODE: $y' = f(t,y)$, $y(0) = \alpha$)

③ $\exists$ constants $\varepsilon_0 > 0$, $k > 0$ such that $\forall \varepsilon$ in $(0, \varepsilon_0)$, whenever $\delta(t)$ is continuous with $|\delta(t)| < \varepsilon$, $\forall t \in [0,T]$ and when $|\delta_0| < \varepsilon$, the initial-value problem $\frac{dz}{dt} = f(t,z) + \delta(t)$, $0 \le t \le T$, $z(0) = \alpha + \delta_0$ has a unique solution $z(t)$ that satisfies $|z(t) - y(t)| < k\varepsilon$ $\forall t \in [0,T]$

Midpoint Rule:	General explicit s-stage RK method:
$\begin{cases} k_1 = f(t_i, y_i) \\ k_2 = f(t_i + \frac{1}{2}h, y_i + \frac{1}{2}hk_1) \\ y_{i+1} = y_i + hk_2 \end{cases}$	$\begin{cases} K_\ell = f(t_i + c_\ell h, y_i + h\sum_{j=1}^{\ell-1} a_{\ell j} K_j) \\ y_{i+1} = y_i + h\sum_{\ell=1}^{s} b_\ell K_\ell \end{cases}$

$$\begin{array}{c|c} c & A \\ \hline & b^T \end{array}$$

1st order accurate: $b_1 + b_2 = 1$; 2nd order accurate: $\begin{cases} b_1 + b_2 = 1 \\ ab_2 = \frac{1}{2} \end{cases}$

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$F(i+1,j+1) = \dfrac{F(i+1,j) - F(i,j)}{X(i+1) - X(i-j+1)}$

Problem 7: Matlab [6 points]

(a) Explain what the following matlab function computes.

Listing 1: MATLAB function to compute ...

```
function F = strangefunction(x, f)
% Input: f is a function handle
% x is a vector with points in an interval in R, x(1)<x(2)<...<x(end)

n = length(x)-1;
F = zeros(n+1,n+1);
F(:,1) = f(x);
for i = 1:n
    for j = 1:i
        F(i+1,j+1) = (F(i+1,j) - F(i,j)) / (x(i+1) - x(i-j+1));
    end
end
```

It's computing the Neville table (extrapolation)! $\to P_{i-j+1:i}$ $P_{i,j,i-1}$

The original formula: $Q_{i,j}(x) = P_{i-j:i}(x) = \dfrac{(x - x_{i-j})Q_{i,j-1}(x) - (x - x_i)Q_{i-1,j-1}(x)}{x_i - x_{i-j}}$

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(b) Add some lines of code to check if the entries of the vector x are distinct and return an error if they aren't. (You don't need to rewrite the whole code, just write between which lines of code of the original code you would insert additional statements)

Between

```
for j=1:i
    if X(i+1)=X(i-j+1):
        print("The entries of the vector X are not distinct!")
        break
    else:
        F(i+1,j+1)= (...)/(...)
end
```

insert

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Space for additional work

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Space for additional work

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Space for additional work

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Space for additional work

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